

Candidate Pseudolabel Learning for Vision-Language Models

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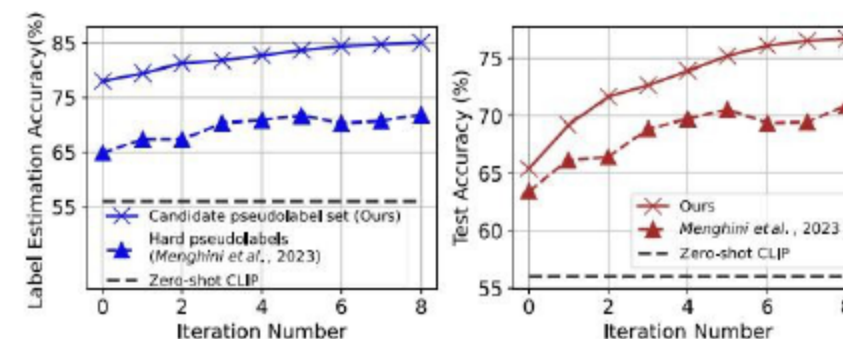
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Motivation & CPL Abstract

3 Abstract & Motivation: Why CPL for VLMs

- Adaptation is label-hungry; unlabeled data is abundant.
- Hard pseudolabels are brittle, especially with low zero-shot.
- CPL keeps a candidate set per sample, training on multiple targets.
 - Confidence matrix guides intra/inter-instance true-label selection.
 - Class-balanced selection boosts few-shot and semi-supervised results.
- Validated across datasets and label regimes with consistent gains.

4 Key Contributions and Early Results



- CPL forms candidate pseudolabel sets, not single.
 - Intra- and inter-instance selection via scores.
 - Confidence matrix ensures true label inclusion.
- Class-balanced candidate selection improves VLMs.
 - Robust training objective reduces noisy labels.
 - Consistent accuracy gains over strong baselines.

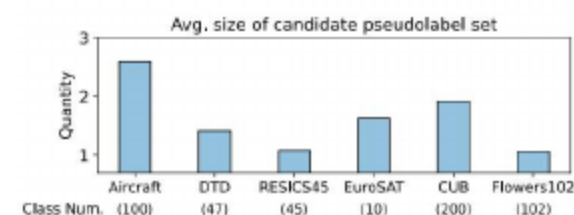
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Related Work and CPL Positioning

1+

CPL vs Prior VLM Adaptation

Methods	Balanced	Imbalanced $\delta=100$	Imbalanced $\delta=50$
LaPTer	74.64	65.63	66.59
CPL (w/o inter)	76.07	66.68	67.85
CPL	77.32	67.70	69.65



- VLMs (CLIP, ALBEF, Flamingo): CPL adapts them without labels.
 - Prompt tuning (text/visual) boosts tasks via efficient updates.
 - Standard pseudolabels pick one; CPL trains on candidate set (>1).
- Balanced vs imbalanced: CPL adapts set size for better labels.

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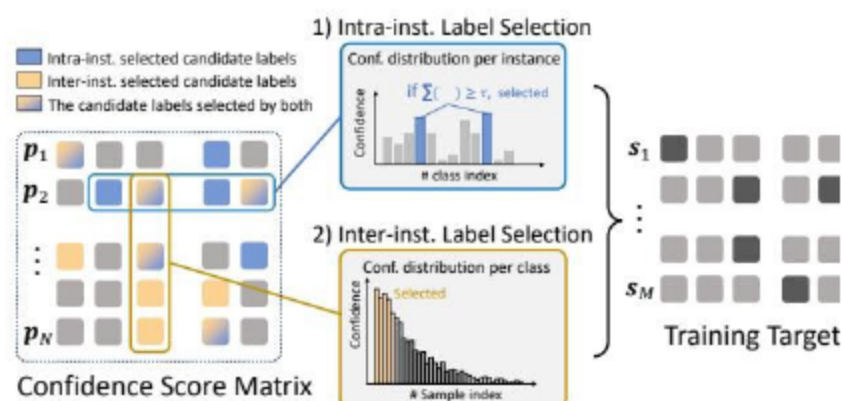
CPL: Setup->Pseudolabels->Train

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Problem Definition and Notation

- Datasets: labeled D_l and unlabeled D_u ; C classes.
- Each class c has a prompt; VLM scores $p(y|x, \text{prompt})$.
- Zero-shot predictions seed candidate pseudolabel sets.
 - Goal: learn prompts/params using unlabeled evidence.
 - Supports SSL, TRZSL, and fully Unsupervised Learning.
- Objective: improve C -class recognition via CPL.

9 Pseudolabel Generation and CPL Training

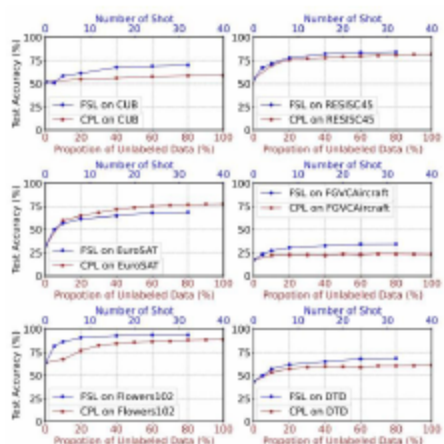


- Intra-instance top-k from confidence scores
 - Inter-instance calibration curbs imbalance
 - Thresholded candidates; final set is intersection
- Candidate-set loss weighted by confidence
 - Jointly trains on labeled and unlabeled data
 - Iterative updates improve accuracy over time

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Experiments & Evaluation

11 Results: Datasets and Label Regimes



- CPL yields consistent gains across six datasets
 - Improvements grow with more unlabeled samples
 - Robust at low-shot regimes and limited labels
- Outperforms FPL and GRIP in weaker zero-shot
 - Scales across semi-, unsupervised, and TRZSL
 - Effective with tuned prompts and hyperparams

12 Ablations, Limitations, and Conclusion

- Ablations: α , β modest sensitivity; loss choice insensitive.
- Tail gains: accuracy rises for rare classes under CPL.
- Confusion cuts: stronger diagonals in matrices with CPL.
 - Limits: relies on prompt quality; candidate search is costly.
 - Similar classes and missing labels can mislead candidate sets.
- Conclusion: candidate-set CPL boosts VLMs across regimes.